Laszlo Anka

Software & Systems Architect | Distributed Systems | Data Platforms | Cloud-Native Infrastructure

Germany | laszlo.anka@gmail.com | linkedin.com/in/laszlo-anka | github.com/lazlowa

PROFILE

Creative and pragmatic problem solver with over two decades of experience in industrial software and systems development. Since 2019, I have specialized in data platforms and cloud-native infrastructure. I combine architectural ownership with hands-on engineering and have previously built and led an engineering organization of more than 30 people. I value honest human relationships and the process of building things.

SELECTED CONTRIBUTIONS

- **Framework development and adoption:** Created and maintain pypz, an open-source Pipeline-as-Code framework that became the business-unit standard and is used by 100+ projects. Reduced typical processing jobs from about a week to around 20 minutes, enabling fast iteration on KPI evaluations for customers.
- **Distributed systems engineering:** Designed and built a data-asset discovery platform connecting 100+ distributed scanning agents to 10+ on-premises databases and local machines, giving users a single interface to find where data actually resides.
- **Developer enablement:** Built a self-service development platform with automated, isolated Kubernetes environments per branch, shortening feature delivery from weeks to a few days through a fully traceable path from branch to production.
- **Organizational development:** Established an artificial intelligence group of 30+ experts from scratch, building both the team and its research and engineering capabilities for frontier topics such as prediction and planning.

SKILLS

- **Systems architecture & engineering:** system decomposition, distributed systems, event-driven services, APIs and interfaces, integration, extensible frameworks
- **Data platforms & engineering:** Kafka, RabbitMQ/AMQP, data ingestion and pipelines, Azure Databricks, Unity Catalog, metadata management, data governance
- **Cloud & infrastructure:** Azure, Kubernetes, Istio, Helm, Docker, Terraform/Terragrunt, cert-manager, monitoring and alerting
- **Software development:** Python, Java/Spring Boot, C/C++ (embedded)
- **Developer platforms & CI/CD:** GitHub Actions, Azure DevOps, monorepos, ephemeral environments
- **Security & identity:** Entra ID, OAuth2/OIDC, workload identity, secrets management
- **Databases:** MongoDB, Neo4j, PostgreSQL
- **AI:** AI-assisted software development, ML training-data workflows, computer vision, AI research leadership
- **Leadership:** engineering team development, recruitment, mentoring, budgeting

PROFESSIONAL EXPERIENCE

Senior Expert, Software and Systems for Data Engineering and Infrastructure

Robert Bosch GmbH, Germany | 06/2019 – present

- **AI-assisted software development & technical problem solving** (10/2026 – present): Contribute to the rollout of AI-assisted coding practices across business units as part of a central enablement team and solve technical problems for units without dedicated developers.
- **Developer platform modernization** (07/2026 – 10/2026): Redesigned the development setup of a pypz pipeline team: established a monorepo, a release process and isolated per-branch Kubernetes deployments automated with GitHub Actions.

- **Cross-company data platform** (02/2024 – 07/2026): Rebuilt the data labeling pipeline from the ground up as event-driven services on Kubernetes (Python/FastAPI, Kafka, MongoDB), replacing an Azure Functions design; it delivers 10,000+ labeling requests per day, each with up to several hundred sensor data samples, to external labeling suppliers with end-to-end traceability and failure tolerance. Provisioned and operated multiple Azure Databricks workspaces using Terraform and built connector services and a secure webhook ingestion API. Worked on silver- and gold-layer processing and data governance with Unity Catalog as part of the governance team.
- **Data-asset discovery platform** (08/2023 – 02/2024): Designed and implemented distributed data-source scanning services, Kafka-based communication and Neo4j-backed metadata storage, consolidating metadata from heterogeneous sources into a graph-based registry (Java/Spring Boot, AKS).
- **pypz adoption and enablement** (03/2023 – 08/2023): Coached a team in building its first processing pipelines with pypz, laying the foundation for its adoption as the business-unit standard.
- **Field-data processing** (06/2022 – 03/2023): Designed continuous data ingestion and preprocessing for data collected from vehicles in real-world operation. Developed an extensible processing framework that lets developers integrate their own code and store results in Databricks. Managed the infrastructure with Terraform.
- **Metadata management** (06/2021 – 06/2022): Built automated ingestion and preprocessing for recorded data feeding a central metadata store.
- **Hadoop integration services** (06/2019 – 06/2021): As a member of the platform team, implemented services connecting the Hadoop platform to internal projects, from requirements analysis through delivery.

Career break and relocation from Hungary to Germany
01/2019 – 05/2019

Robert Bosch Kft., Budapest, Hungary
09/2007 – 12/2018

- **Section Manager, Artificial Intelligence** (2016 – 2018): Built the AI group from scratch, including recruitment; established research and infrastructure capabilities, and partnerships with the global Bosch AI organization and Hungarian universities and institutes (ELTE, BME, MTA SZTAKI).
- **Section Manager, Computer Vision** (2015 – 2018, concurrent): Built a 15-person computer vision team and coordinated the site's computer vision activities with product teams in Germany.
- **Senior Expert, Computer Vision** (2013 – 2015): Led the technical direction of all computer vision activities at the site.
- **Computer Vision Architect, Leonberg, Germany** (2012 – 2014): On-site assignment in Germany; owned the vision software architecture of a first-generation product, including resource allocation, logging and measurement infrastructure.
- **Image Processing Algorithm Developer** (2007 – 2012): Developed camera/radar sensor fusion and monocular distance estimation algorithms in C/C++.

Internship and part-time role
HSG-IMIT Research Institute, Villingen-Schwenningen, Germany | 09/2006 – 09/2007

- Developed hardware and software for a voice-controlled, glasses-mounted camera system.

OPEN SOURCE

- **pypz, creator and maintainer** (github.com/lazlowa/pypz-python, pypz.dev): Python Pipeline-as-Code framework for parallel, scalable data processing, with Kafka and RabbitMQ plugins, Kubernetes deployment and automated releases to PyPI and Docker Hub (Apache 2.0).

EDUCATION

- **MSc Biomedical Engineering**, Furtwangen University, Germany (2005 – 2007)
- **BSc Electrical Engineering**, Budapest Tech (BMF), Hungary (2001 – 2005); laboratory assistant (2004 – 2005)

LANGUAGES

Hungarian (native), German (fluent), English (fluent)